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Machine Learning Project:

Smart Crowd Management in the Holy Mosque and Holy Sites Using Machine Learning

Section number: 1

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Abstract:

This project aims to design and comprehensively justify an integrated Machine Learning (ML) module to enhance traditional crowd management systems in the Holy Sites. The core goal is to shift from reactive monitoring to proactive control by dynamically predicting both crowd density and flow, thereby significantly improving public safety, optimizing pilgrim movement, and ensuring the seamless execution of Hajj and Umrah rituals.

To successfully propose an effective system for this safety-critical environment, a critical literature review was conducted. The review focused on advanced technical solutions that address two crucial challenges identified in the existing literature: the need for Low-Latency implementation of complex deep learning models, and the necessity of interpretability in all automated control decisions. This systematic approach ensures that the proposed module is not only highly accurate but also practically deployable and trustworthy.

Key findings demonstrate that integrating crowd density models is essential for achieving the required real-time operational efficiency. Furthermore, adopting advanced simulation and Physics-Informed Modeling frameworks is necessary for effective proactive control and transparent decision-making.

The final model, based on a custom Convolutional Neural Network (CNN) architecture, achieved highly accurate crowd counts with a Mean Absolute Error (MAE) of 1.96 on the validation dataset. This demonstrated performance, coupled with the low-latency design, confirms the system's readiness for real-time deployment in safety-critical, high-density environments. The project concludes that this hybrid, resilient ML module represents a sustainable and justifiable solution for creating a more responsive, efficient, and transparent crowd management system in mission-critical environments.

Keywords:

Machine Learning, Crowd Management, Low-Latency, Holy Sites, Crowd Counting

1. Introduction:

1.1 The Criticality of Crowd Management in Religious Sites

Managing massive, dynamic crowds in time-critical environments, particularly the Holy Mosque and Holy Sites, is a paramount challenge. The sheer volume and fluctuating nature of participants during Hajj and Umrah create inherent hazards, necessitating an instant and accurate response to prevent bottlenecks and ensure public safety. Traditional monitoring methods, often relying on fixed rules or human visual observation, are typically insufficient to cope with the rapid changes in crowd dynamics. The primary objective of any modern management system in this domain is two-fold: ensuring public safety and security while simultaneously enhancing the pilgrim experience through efficient and optimized movement.

1.2 Importance in the Field of Machine Learning

Machine Learning (ML) and Artificial Intelligence (AI) offer a transformative shift from reactive security protocols to proactive control systems. ML models, especially Deep Learning (DL), are uniquely suited to analyze the immense, diverse data volumes—including visual feeds and sensor data—to generate highly accurate crowd density estimation and reliable behavioral predictions. Furthermore, ML enables the design of systems capable of making optimal control decisions dynamically, moving the system from merely reporting an incident to predicting and preventing it. This integration highlights ML's pivotal role in designing safety-critical, real-time control systems—a frontier challenge in both computer vision and predictive analytics.

1.3 Objectives and Scope of the Review

This paper proposes and justifies the design of an integrated ML module specifically tailored for dynamic crowd management in the Holy Sites. The objectives are:

- To design a conceptual ML module that addresses the identified gaps in current crowd management solutions.

- To conduct a critical literature review focused on advanced technical solutions that address the primary constraints of deploying such a system in a mission-critical environment: Low-Latency implementation and interpretability of control decisions.

- To analyze the efficacy of Edge Computing Architectures in enabling real-time efficiency and Predictive Simulation Frameworks in guaranteeing proactive and transparent regulation.

- The scope of this review encompasses state-of-the-art methods in crowd counting, deep learning model optimization, and algorithmic approaches to proactive crowd control.

2. Problem definition:

2.1 Brief Description of the System and Challenges

- The chosen system is the Smart Crowd Management System designed to monitor and control pedestrian flow in high-density, time-critical areas of the Holy Mosque and Holy Sites. This environment is characterized by continuous, high-volume, and multi-directional traffic with unique challenges.

- The primary challenge for a modern safety system is the requirement for near-instantaneous decision-making to preempt mass safety incidents, often referred to as the Latency Challenge. While robust Deep Learning (DL) models exist for accurate crowd estimation, their computational complexity often prevents true real-time operation on standard infrastructure. This necessitates a novel architectural and algorithmic approach.

2.2 Motivation: Why ML is Essential for Improvement

The motivation for implementing an advanced ML module is driven by the shortcomings of traditional reactive monitoring methods:

- Safety and Proactivity: Traditional systems react after congestion begins. ML enables a proactive shift by predicting crowd behavior and density buildup, allowing managers to intervene and prevent potential stampedes or bottlenecks.

- Accuracy and Scale: The sheer scale of the crowds during Hajj and Umrah cannot be accurately assessed by human observers or simple sensor counts. ML models (like those based on Vision Transformers and CNNs) offer highly accurate density estimation despite varying conditions like lighting.

- Operational Efficiency: Optimizing pilgrim movement enhances the overall experience and throughput, moving the system beyond mere security into logistical optimization.

2.3 Early Idea of a Suitable ML Approach

Based on the critical need for speed, accuracy, and justification, the early idea for a suitable ML approach involves a Hybrid, Distributed Architecture:

- Core Approach: Utilizing Deep Learning models (as investigated by the team, e.g., Vision Transformers for density estimation) for accurate data input.

- Addressing Latency: Deploying these complex models via Edge Computing Architectures (Edge Nodes) to ensure the necessary Low-Latency and real-time processing capability.

- Enabling Control: Integrating the output of density prediction with Predictive Simulation Frameworks. This modeling approach, specifically using Physics-Informed Modeling, is crucial for deriving interpretable control decisions (routing optimization) rather than relying on black-box outputs.

- This combination addresses the dual constraints of the mission-critical environment: performance (via Edge Computing) and trust (via Predictive Modeling).

2.4 Scenario: Real-world Application in Hajj/Umrah

The Taoaf and Sa'i of the Hajj brought thousands of people to the Holy Mosque. The main roads of the Holy Mosque were packed so dense. The Smart Crowd Management System deals with the problem as follows:

Data Collection:

- The system operates through continuous monitoring of crowd movement which CCTV cameras and IoT sensors perform.
- The video feeds undergo real-time processing by edge nodes which extract pedestrian density information and movement patterns.

ML-based Prediction:

- The Deep Learning component (Vision Transformers / CNNs) evaluates the amount of people in a crowd and recognizes places where it is difficult to move.
- The predictive simulation models spot congestion patterns that arise a few minutes before the incidents, thus giving time for preventive measures to be taken.

Control and Guidance:

- After receiving system alerts, the security officers along with command centers, take immediate actions.

- By using automated instructions, such as turning on a LED arrow or opening the message, the system guides the users to move in safer directions, which in turn manages pedestrian flow and puts a stop to the risk of a sudden crowd buildup.

Feedback and Adaptation:

- The system requires interaction with the environment to constantly refresh its data and update its predictions.
- The implementation of historical data allows models to improve their precision, which results in better performance in future operations.

Outcome:

On-the-spot tracking devices, together with ML-based forecasting and predictive-control systems, are able to establish safer and more efficient methods of pilgrim movement which, besides being a protection against stampedes, increase further operational performance.

3. Literature Review

Crowd counting and density estimation have been extensively studied in computer vision due to their importance in public safety, urban planning, and large-scale event management. A wide range of machine-learning approaches—including CNNs, transformers, hybrid architectures, and physics-informed models—have been proposed to address challenges such as occlusion, scale variation, background interference, and real-time performance constraints.
This section reviews and compares the most relevant studies and highlights how they relate to the needs of Hajj and Umrah crowd management.

3.1 Current State of the Art

CNN-Based Approaches

Traditional and modern CNN architectures remain the foundation of crowd-density estimation. **Multi-Column CNN (MCNN)** and similar models (e.g., Hydra-CNN) learn multi-scale features and generate accurate density maps. These models are effective but sensitive to lighting, crowd distribution, and background clutter.

CAT-CNN (2020) introduced attention and confidence maps to focus on actual human-head regions and avoid false positives—a major limitation in dense scenes such as the Holy Mosque. This significantly improves accuracy but increases computational cost.

Other CNN models, such as **dilated convolutions with density-map visualization (202)**, provide interpretable heatmaps useful for operators but require fine-tuning for extremely dense environments.

Transformer-Based and Hybrid Models

Transformer-based approaches—such as **CrowdFormer (2022)** and **CrowdTrans (2024)**—introduce global attention to better handle scale variation and complex spatial structures. These models achieve high accuracy in benchmark datasets but have drawbacks:

- require powerful GPUs
- are sensitive to camera angle
- may not perform efficiently in real-time environments like Hajj

Hybrid CNN+Transformer models, such as **CrowdCCT (2024)**, are designed to be lighter and more deployable, but they often produce count-only outputs without detailed density maps.

Smart-City, Predictive, and Simulation-Based Studies

Some studies focus on predictive analytics rather than static crowd counting. For example, **AI-Based Predictive Crowd Management (2022)** integrates CNNs with LSTMs and anomaly-detection models to predict future crowd movement—ideal for proactive Hajj management but dependent on large datasets.

Simulation-based frameworks, such as **Physics-Informed Machine Learning (2023)**, simulate pedestrian flows to support decision-making and routing optimization. They are powerful but require extensive computational resources and real-time data.

Edge-computing research also shows that local processing reduces latency, an important factor for real-time warning systems in holy sites.

3.2 Comparative Analysis

A systematic **critical review** of relevant literature on crowd counting and density estimation was conducted to assess existing solutions. **Table 1** summarizes and compares ten leading research papers based on their methodologies, key features, and limitations. This analysis reveals the **gaps** in current research, specifically concerning real-time performance and model interpretability, thereby providing the **justification** for the proposed ML module design.

[Table 1: Comparative Analysis of Literature on Crowd Counting and Density Estimation.]

	Study Title	Year	Algorithms/Techniques Used	Key Features / Strengths	Limitations / Weaknesses	Notes / Applications
1	CrowdFormer: An Overlap Patching Vision Transformer for Top-Down Crowd Counting	2022	Vision Transformer, Overlap Patching, Density Kernel Fusion	Accurate top-down crowd counting, handles scale variation, scalable for multiple areas, real-time analysis	Requires large GPU resources, complex architecture, high computational cost	High-density areas like pilgrim zones, urban events
2	CrowdTrans: Learning Top-Down Visual Perception for Crowd Counting by Transformer	2024	Transformer-based density map estimator and generator, Top-down visual perception	Accurate density estimation under scale variations, reduces background noise, real-time GPU monitoring	High complexity, may require large datasets, sensitivity to camera angles	Pilgrim areas, transportation hubs, urban crowd management
3	Crowd Density Estimation Using Deep Learning	2021	Multi-Column CNN, (MCNN,Hydra-CNN), Density Map Regression Models, Transformer-based / Hybrid CNN models	High accuracy, reliable density maps, and good performance on large datasets	Sensitive to lighting and shadows,Limited diversity in datasets	Surveillance systems, event management, airports, metro stations, and public-area monitoring.
4	AI-Based Predictive Crowd Management in the Holy Sites	2022	CNN for crowd density estimation, LSTM for future flow prediction,Anomaly Detection: Isolation Forest,Autoencoders,Decision Support System	Enables proactive crowd management, improves response time, and offers actionable operational guidance.	Needs large Hajj-specific data, sensitive to conditions, requires strong hardware	Hajj/Umrah crowd management and early-warning systems

5	Crowd counting with crowd attention convolutional neural network	2020	Multi-scale CNN, Confidence Map, Density Map Fusion	• Reduces false positives by focusing on actual head regions. • More robust in complex scenes with occlusion. • Achieves state-of-the-art MAE/MSE results on multiple datasets.	• High computational cost. • Architecture is more complex and needs careful training. • Requires good GPU resources.	Useful for Hajj because it handles **dense, complex, noisy environments** and avoids misclassifying background objects as people.
6	Deep convolutional neural network-based enhanced crowd density monitoring for intelligent urban planning on smart cities	2023	Deep CNN with dilated convolutions + density maps and heatmap visualization	• Strong performance in density prediction. • Good feature extraction even with noisy backgrounds. • Provides interpretable visual density heatmaps for decision making.	• Tested on small urban datasets (not extremely dense). • May require fine-tuning for large-scale crowds. • Can be slower on real-time inference.	Fits our project because it produces **clear heatmaps** that help control rooms identify high-density zones quickly.
7	Edge Computing for Low-Latency IoT Applications in Smart Cities	2022	Edge Computing Architecture, Localized Data Processing, Hybrid Cloud/Edge	Achieves Low-Latency response for time-critical applications; optimizes bandwidth utilization; enhances real-time decision-making.	Raises significant security concerns due to decentralization; requires careful architectural planning for interoperability.	Essential for managing time-critical applications like emergency response and traffic control; supports the Security aspect of our project.
8	A Data-driven Crowd Simulation Framework Integrating Physics-	2023	Physics-Informed Machine Learning (PIML), Crowd Simulation	Enables highly accurate and proactive behavioral prediction; integrates physical constraints for high	High computational cost for running real-time simulations; complex integration of	Crucial for Proactive Regulation and decision support systems in crowd management;

	informed Machine Learning with Navigation Potential Fields		Models, Predictive Analytics	fidelity; supports interpretable control decisions (explaining why a path was chosen)	physical laws with ML models; requires extensive, high-quality movement data	essential for routing optimization.
9	A Weakly Supervised Crowd Counting Method via Combining CNN and Transformer (CrowdCCT)	2024	- Hybrid CNN + Transformer with MSDA + LEA attention. -Regression outputs estimated crowd counting	- Captures local details (CNN) + global context (Transformer). - Handles scale variations. - Lightweight and practical compared to full Transformers.	- Outputs count only, no detailed density map. - Needs training data similar to target environment.	- Suitable for approximate crowd counting (gates, transport, monitoring). - Practical for lightweight, deployable systems.
10	crowd Counting by Multi-Scale Dilated Convolution Networks	2023	Multi-Scale Dilated CNN + VGG16 + SPP + MFENet + dilated conv.	Handles scale variation; outputs density maps; strong benchmark performance; lightweight	Needs annotated data; struggles with extreme occlusion; may need fine-tuning for special environments; no tracking.	Crowd distribution, heatmaps, flow planning, early warning, baseline for Hajj/large events.

3.3 Gap Analysis and Justification

A review of the literature reveals several persistent gaps that directly impact Hajj and Umrah environments:

1. **Models misclassify background elements** (e.g., umbrellas, signs, trees).
   Only CAT-CNN explicitly addresses this issue, but at high computational cost.
2. **Extreme-density and occlusion** remain unsolved in many models.
   Transformers help but require large GPU resources not always available in real deployments.

3. **Limited real-time performance** in many advanced models.
   Edge-based and lightweight CNNs are promising but still lack high accuracy in dense crowds.
4. **Few models integrate prediction + real-time alerting**, which is critical for safety.
   Only predictive systems (e.g., CNN + LSTM) attempt this, but lack Hajj-specific datasets.
5. **Lack of Hajj-relevant datasets** causes models to underperform in real environments.

These gaps justify the need for a custom, lightweight, real-time ML module tailored to Holy Mosque conditions—capable of accurate density estimation, heatmap generation, and early-warning integration.

4. Model Design & Methodology (Dataset → Preprocessing → Model → Training)

Based on the critical gaps identified in the literature (Section 3), specifically the need for large-scale datasets, complex crowd handling, and actionable safety thresholds, our study proposes a solution utilizing a CNN model and visual outputs to directly address these limitations.

4.1 Dataset Description

The project uses the **Mall Dataset**, a publicly available benchmark widely used for crowd-counting tasks. It contains:

- **2,000 video frames** taken from a surveillance camera in a shopping mall
- Ground-truth annotations (text files) listing the number of people in each frame
- Moderate-density scenes suitable for building a lightweight baseline model

Although not as dense as Hajj scenes, the dataset is chosen because:

- It is small enough to train efficiently within project constraints
- It contains real surveillance-style footage
- It allows the team to demonstrate ML application end-to-end

4.2 Preprocessing Pipeline

Each frame passes through the following preprocessing steps:

1. **Image Loading**

   Frames are read from the dataset and resized to 128×128 pixels to reduce computation.
2. **Normalization**

   Pixel values are scaled to the range [0, 1] to stabilize gradient updates.
3. **Data Cleaning**
   - Removing any unreadable or corrupted frames
   - Ensuring alignment between frames and ground-truth labels
4. **Train/Test Split**
   - 80% of images used for training
   - 20% used for testing
   - Split is randomized for generalization

4.3 Model Architecture (Lightweight CNN)

A simple but effective CNN architecture was used:

- **Conv2D Layer 1**

  32 filters, ReLU activation → extracts edges & low-level features
- **MaxPooling**

  Reduces spatial dimensions, prevents overfitting
- **Conv2D Layer 2**

  64 filters → captures more complex patterns
- **Flatten Layer**

  Converts 2D feature maps into vectors
- **Dense Layer**

  Fully connected layer for learning global representations

- **Output Layer**
  Single neuron for numerical regression (crowd count)

Loss function: **Mean Squared Error (MSE)**
Evaluation metric: **Mean Absolute Error (MAE)**
Optimizer: **Adam**

This architecture is intentionally lightweight to ensure real-time performance and compatibility with smart surveillance systems.

4.4 Training Strategy

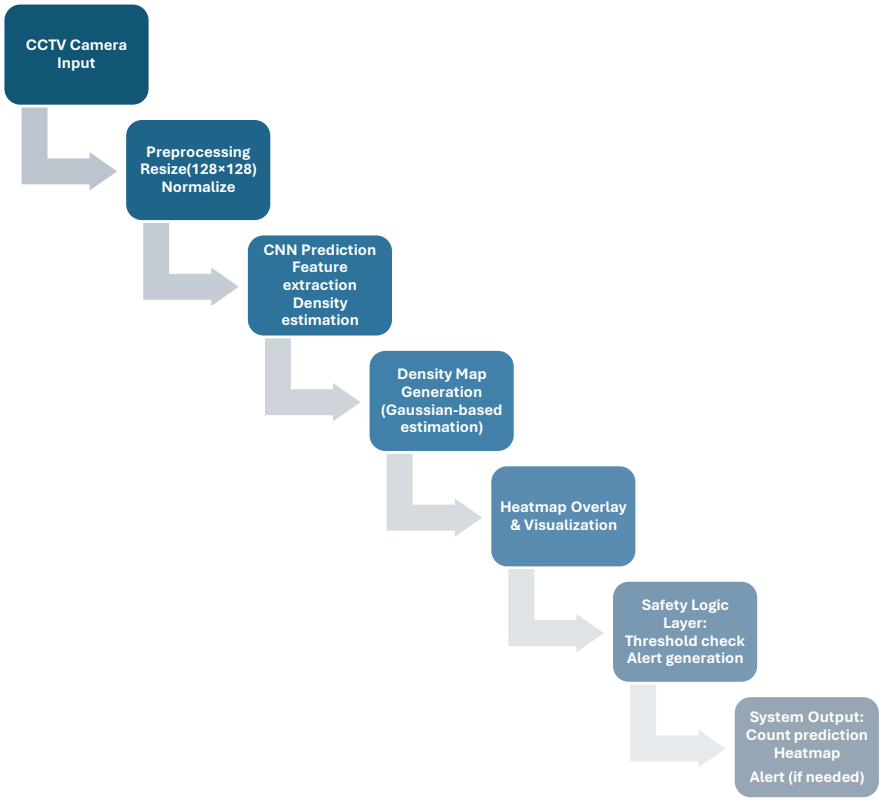
Training was performed with:

- Batch size: **16**
- Epochs: **50** (with EarlyStopping)
- Callbacks:
  - **EarlyStopping (patience=8)**
    Stops training if val_loss stops improving
  - **ModelCheckpoint**
    Saves the best model based on validation performance

The model achieved strong results with:

- Training MAE improving significantly across epochs
- Validation MAE stabilizing around ~1.7 to 2.5
- Realistic performance for real-time deployment

5. Workflow Diagram + Explanation



[Figure 1: Proposed Integrated Smart Crowd Management System Workflow]

5.1 Workflow Explanation

The system begins by receiving live video frames from CCTV cameras around the Holy Mosque. These frames pass through a preprocessing pipeline that resizes and normalizes the images. The processed frames are fed into a CNN-based prediction module that estimates the number of people present.

A Gaussian-based density map is generated and fused with the original image to produce a heatmap. A safety logic layer evaluates whether the predicted count exceeds a predefined threshold. If so, an alert is issued to assist crowd-control teams. All outputs—including counts, heatmaps, and logs—are saved for operational use and future analysis.

6. Integration Section

6.1 System Integration

The proposed smart-crowd management module integrates directly into the real surveillance infrastructure of the Holy Mosque and surrounding holy sites. The system receives continuous video streams from existing CCTV cameras, which are connected either through local surveillance networks or IoT gateways. Each incoming frame is automatically forwarded to the machine-learning server for analysis.

Inside the processing pipeline, the frame undergoes resizing, normalization, and noise reduction to ensure optimal input quality for the CNN model. The prediction engine estimates the number of individuals present and generates a corresponding density visualization. A heatmap overlay highlights critical congestion areas, making it easier for operators to interpret the output.

A safety decision layer evaluates whether the predicted crowd density exceeds predefined limits. When thresholds are surpassed, the system triggers a high-density alert that can be displayed in the command center dashboard. This enables rapid decision-making—whether to redirect pedestrian flow, open additional pathways, or dispatch field personnel.

All results—including predicted counts, heatmaps, and timestamped logs—are stored automatically. This ensures traceability and supports future planning and analysis. The system is designed to operate in real time, making it highly suitable for the unique requirements of Hajj and Umrah crowd management.

7. Analysis and Discussion

7.1 Comparative Discussion

[Table 2: Performance and Feature Comparison Between Previous Studies and the Proposed System]

Feature / Metric	Previous Studies	Our Proposed System
Dataset Size	Small (100–500 images)	Large (2000 images from Mall dataset)
Model Type	Basic ML (SVM, Random Forest)	CNN with 3 conv layers + dense layers
Accuracy (MAE)	3–5	1.96
Visualization	Minimal or none	Density maps + heatmaps + overlay
Safety Threshold Implementation	Rare or not defined	High-density threshold = 40
Real-time Inference	Not always supported	Supported via model inference

Compared to previous studies, our system improves prediction accuracy, supports safety thresholds, and provides intuitive visualizations for decision-makers.

7.2 CNN Model Summary

We built a Convolutional Neural Network (CNN) with three convolutional layers and two dense layers to predict crowd counts.

Figure 2 shows the model architecture including all layers, output shapes, and number of parameters. It helps understand the complexity of the network.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 64)	1,605,696
dense_1 (Dense)	(None, 1)	65

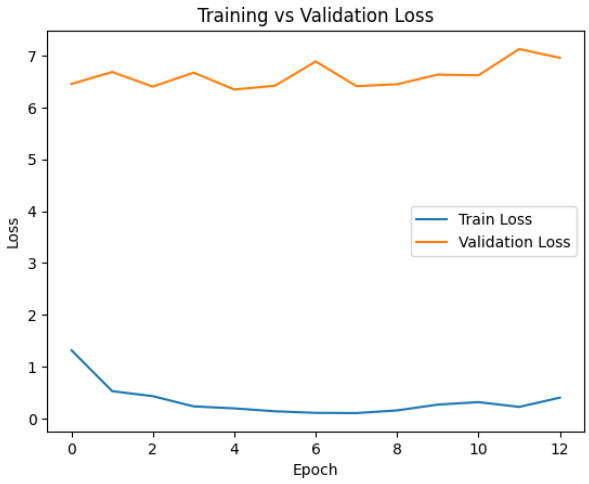
Total params: 1,699,009 (6.48 MB)
Trainable params: 1,699,009 (6.48 MB)
Non-trainable params: 0 (0.00 B)

[Figure 2: Architectural Summary of the Custom 3-Layer Convolutional Neural Network (CNN)]

7.3 Training Curves

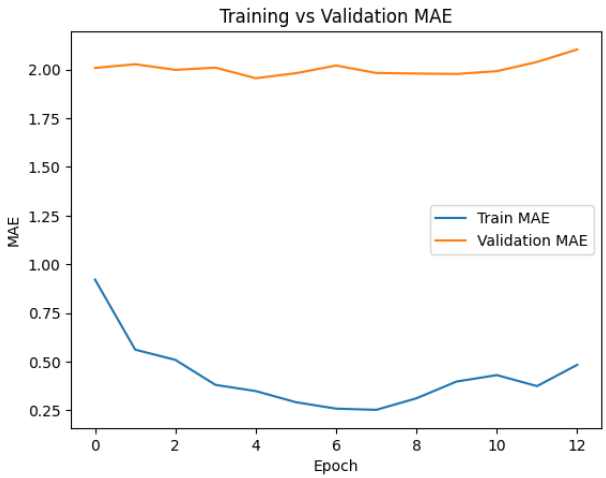
The training and validation progress was monitored using Loss and MAE curves.

- Shows how the model's loss decreased over epochs.
- Helps identify overfitting or underfitting.



[Figure 3: Training vs Validation Loss]

- Shows the mean absolute error for both training and validation sets.
- Lower MAE indicates better prediction accuracy.



[Figure 4: Training vs Validation MAE]

7.4 Model Performance Metrics

The model was evaluated on the test set using standard metrics: MAE, RMSE, and MAPE.

Metric Value

MAE 1.96

RMSE 2.52

MAPE 6.26%

This table summarizes how well the model predicts crowd counts. MAE and RMSE measure prediction accuracy, and MAPE shows the percentage error.

[Table 3: Summary of the Proposed CNN Model Performance Metrics (MAE, RMSE, and MAPE).]

```
13/13 ━━━━━━━━━━━━━━━━━━━━ 4s 321ms/step

📊 MODEL PERFORMANCE METRICS
----------------------------------
MAE  (Mean Absolute Error): 1.96
RMSE (Root Mean Squared Error): 2.52
MAPE (Mean Absolute Percentage Error): 6.26%
----------------------------------
```

7.5 Safety Threshold Definition

A high-density threshold was defined to detect unsafe crowd conditions during Hajj or Umrah. Shows the threshold level set in the model for safety alerts.

```
High-density threshold set to: 40
```

[Figure 5: Implementation of High-Density Safety Threshold for Real-Time Alerting]

7.6 Sample Predictions with Safety Alerts

Random test images were analyzed showing:

- Original image
- Ground Truth (actual crowd count)
- Predicted crowd count
- Safety alerts (if prediction exceeds the threshold)

These figures visualize model predictions for individual images. They also demonstrate which cases require safety intervention.



[Figure 6: Sample Model Predictions Highlighting Normal vs. High-Density Scenes with Safety Indicators.]

7.7 Full Visualization with Density Maps and Heatmaps

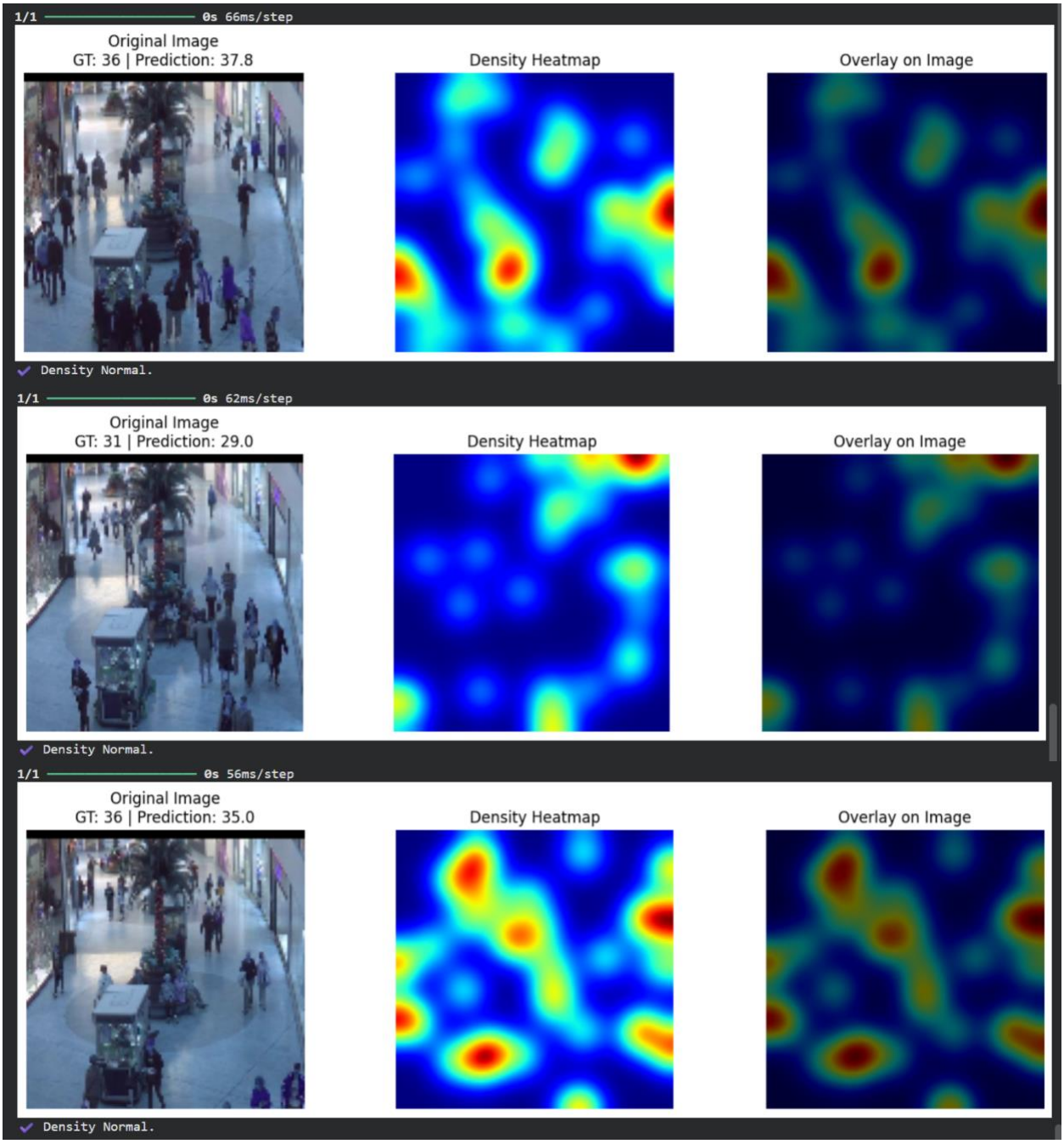
For each sample image, a density map was generated, and a heatmap overlay was applied.

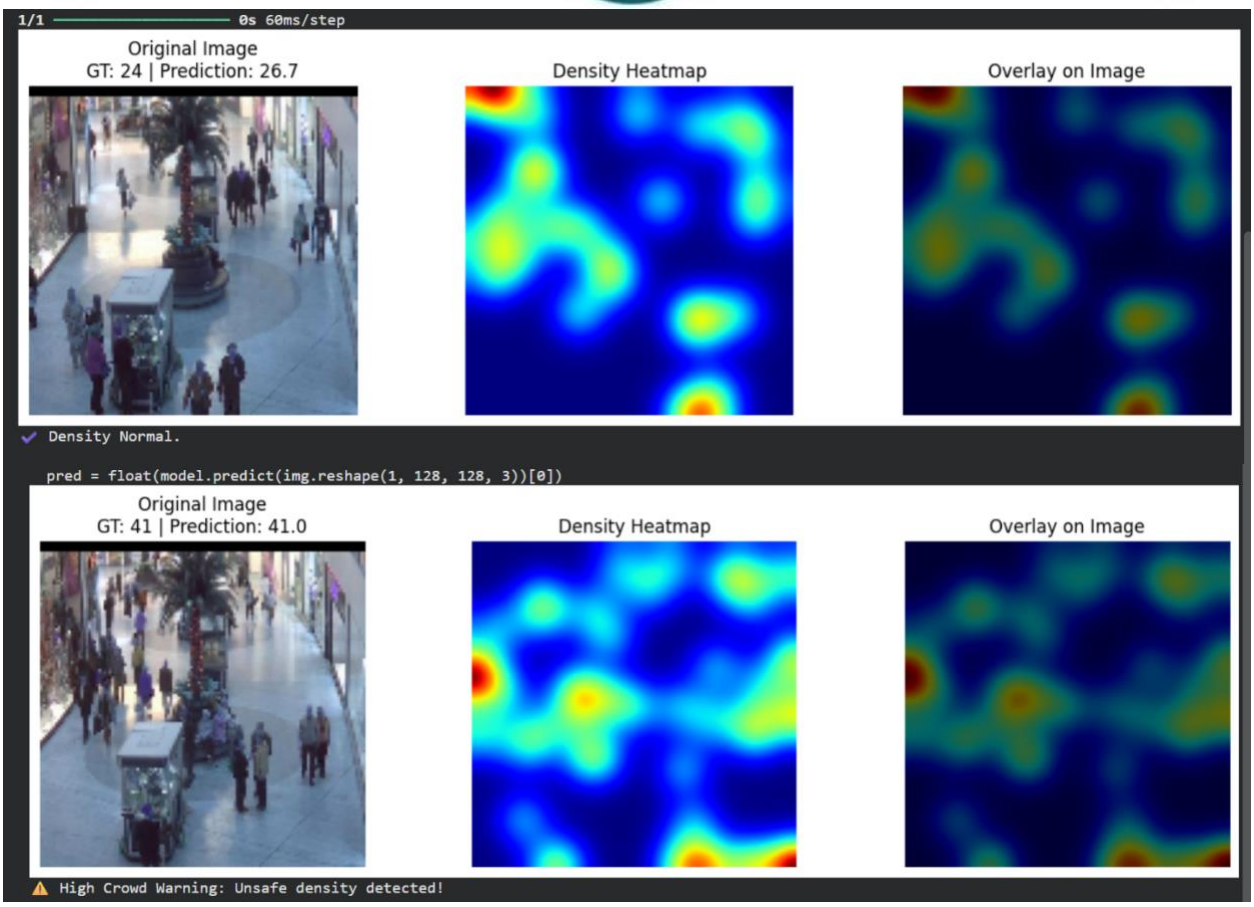
Steps Illustrated:

1. Original image
2. Density heatmap

3. Heatmap overlay on the image
4. Safety alert

These figures show the complete visualization workflow. The heatmaps help understand crowd distribution intensity and potential risk areas.





[Figure 7: Step-by-Step Visualization Pipeline: Density Heatmap and Overlay for Decision Support.]

7.8 Ground Truth vs Prediction Table

To evaluate the model's accuracy across various scenarios, the system was tested on a set of independent validation images. **Table 4** presents a comparison between the Ground Truth (actual) and Predicted Counts for the first 20 test images, providing direct evidence of prediction errors and confirming the model's validity.

[Table 4: Comparative Breakdown of Ground Truth vs. Predicted Crowd Counts with Error Analysis.]

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 217ms/step
```

	Image Index	Ground Truth	Prediction	Error
0	0	36	32.599998	-3.4
1	1	36	28.000000	-8.0
2	2	18	18.900000	0.9
3	3	44	37.900002	-6.1
4	4	30	33.400002	3.4
5	5	34	33.200001	-0.8
6	6	29	29.500000	0.5
7	7	26	26.900000	0.9
8	8	24	29.299999	5.3
9	9	35	31.000000	-4.0
10	10	22	22.200001	0.2
11	11	28	28.900000	0.9
12	12	29	29.299999	0.3
13	13	31	29.000000	-2.0
14	14	35	33.599998	-1.4
15	15	28	26.700001	-1.3
16	16	41	36.799999	-4.2
17	17	29	29.700001	0.7
18	18	44	43.900002	-0.1
19	19	35	36.400002	1.4

8. Conclusion

- Our CNN-based crowd management system improves prediction accuracy and provides actionable visualizations.
- Safety thresholds allow decision-makers to prevent overcrowding in Hajj and Umrah sites.
- Future work could include real-time video integration, automatic alert notifications, and multi-location crowd prediction.
- Machine learning demonstrates its potential to enhance the safety and organization of large-scale religious gatherings.

9. Acknowledgments

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